CSEP 590tv: Quantum Computing

Dave Bacon

Aug 17, 2005

Today's Menu

Quantum Computing implementations

Quantum Error Correction

Quantum Cryptography

Quantum Information and Black Holes…

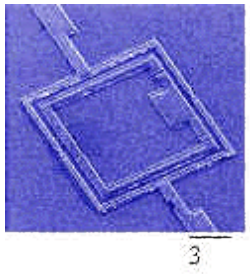
Administrivia

Turn in the take home final. Let out a deep breath.

If you are taking the 1 week extension which is an extension to Monday, please let me know via email.

Fill out course evaluations at end of class.

But What Will It Look Like?



Solid State

superconducting circuits

electron spin in Phosphorus doped Silicon

quantum dots

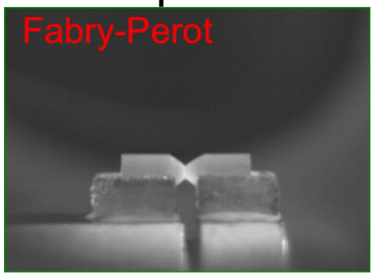
defects in diamonds

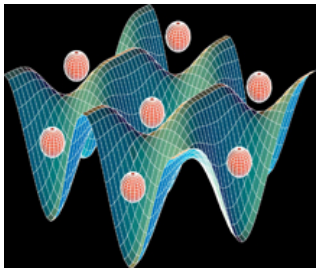
Atomic

cavity QED

neutral atoms in optical lattices

ion traps





Molecular

Liquid NMR (no longer?)

Photon Based

linear optics plus single photon devices

Pics: Mabuchi (Caltech), Orlando (MIT)

DiVincenzo's Criteria



David DiVincenzo

1. Well defined qubits in a scalable architecture

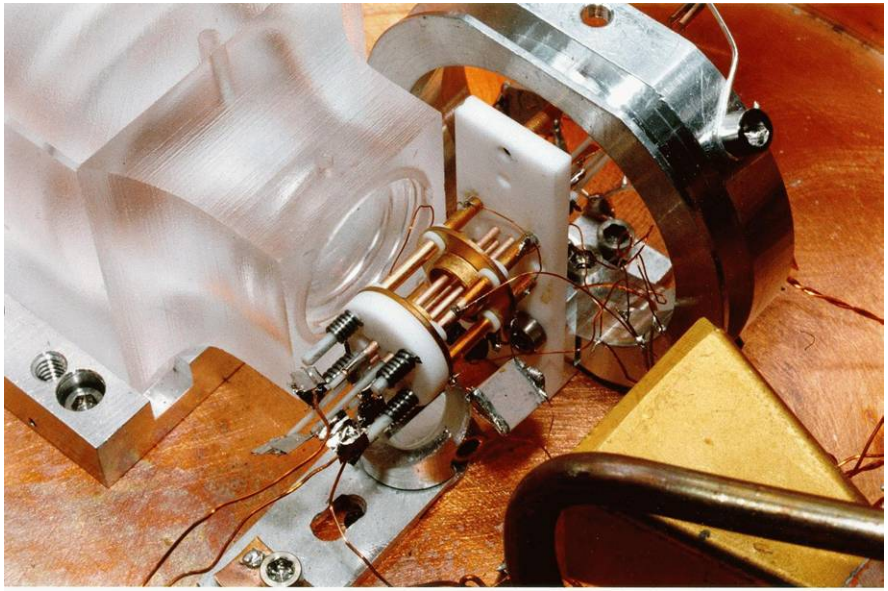
2. The ability to initialize the system to a fixed wave function.

3. Have faster control over the system than error processes in the system.

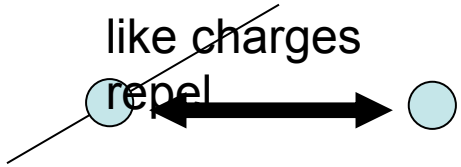
4. Have the ability to perform a universal set of quantum gates.

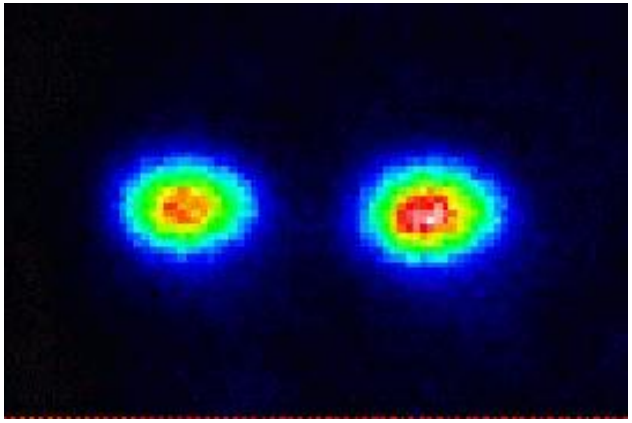
5. Have the ability to perform high quality measurements

Ion Trap

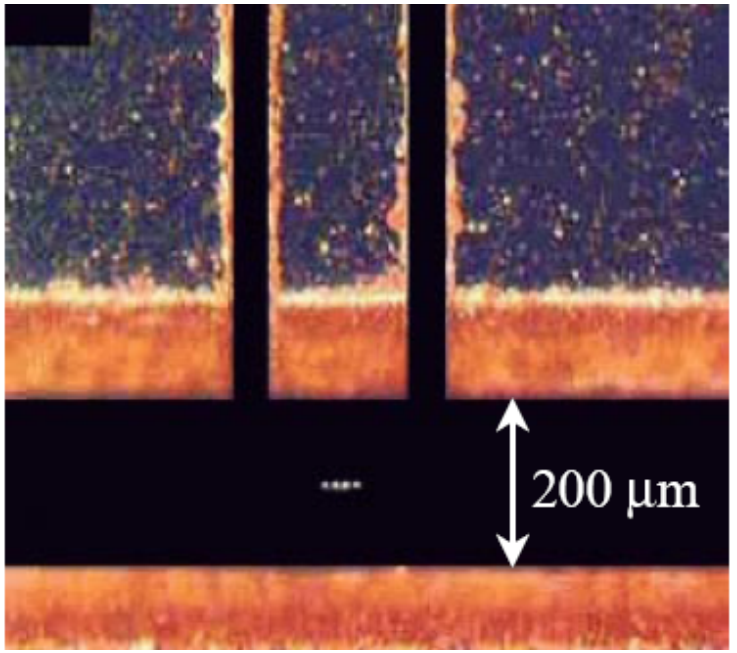


Oscillating electric fields trap ions

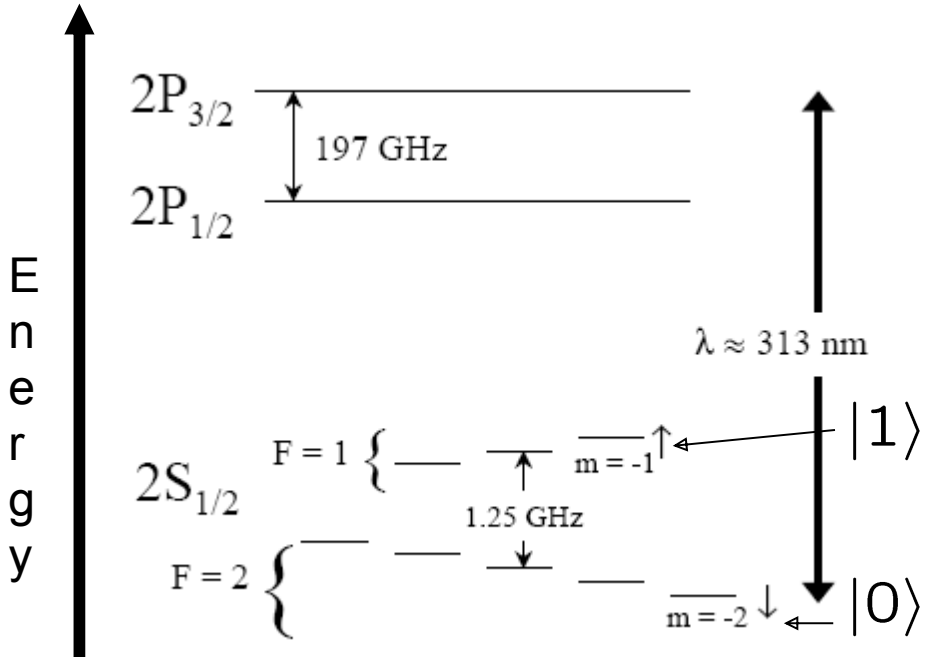


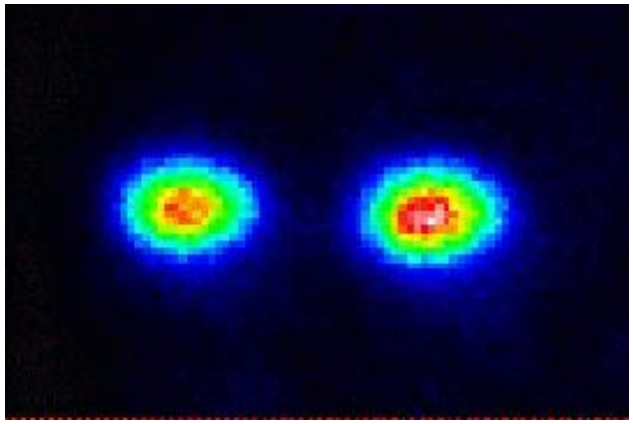


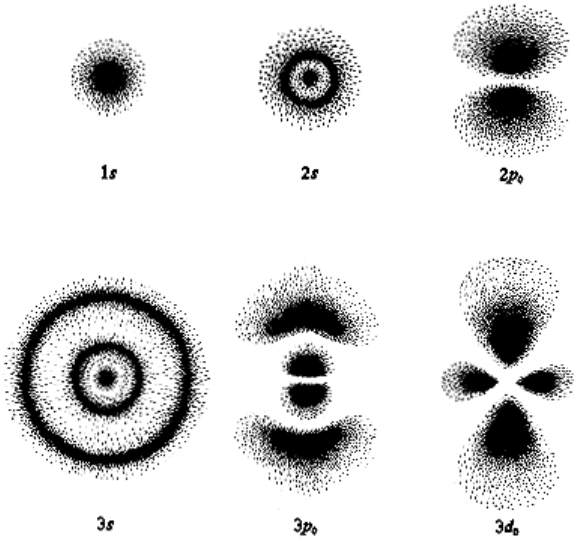
2 9Be+ Ions in an Ion Trap



Where's the Qubit?





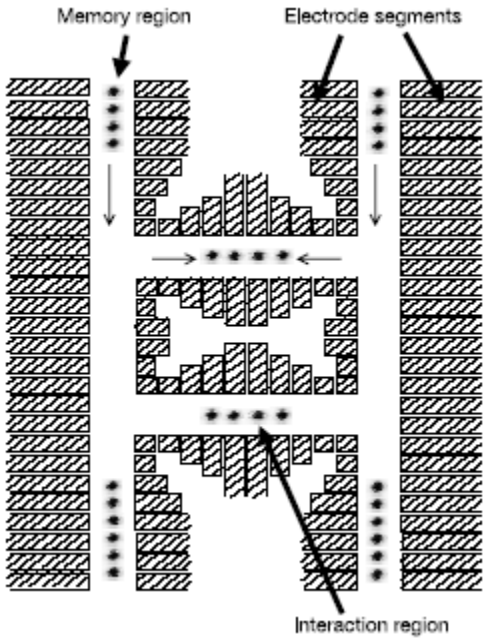


Each ion = 1 qubit

1. Well defined qubits ♠

orbitals

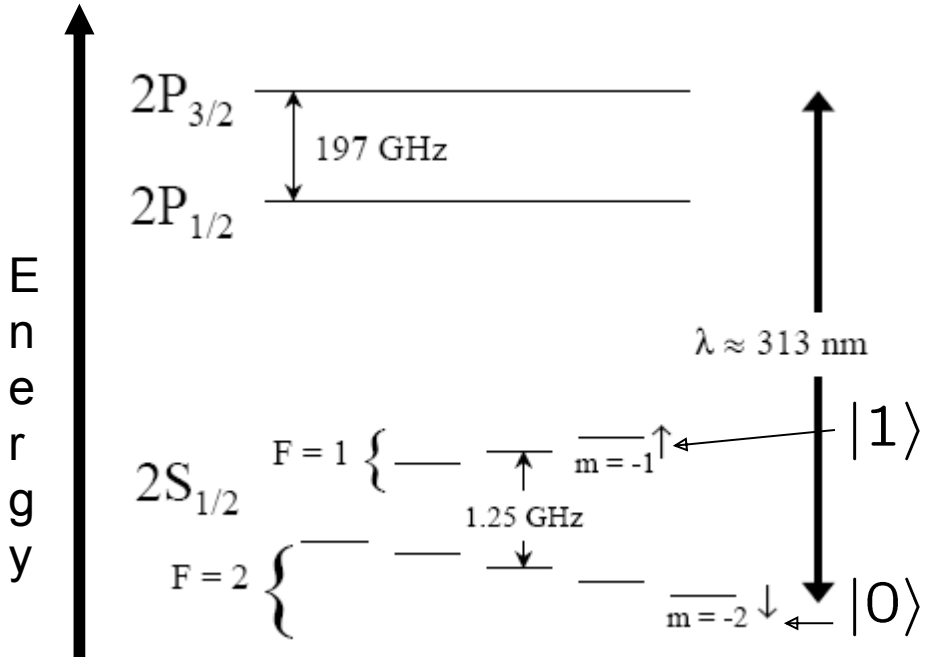
Scalable?

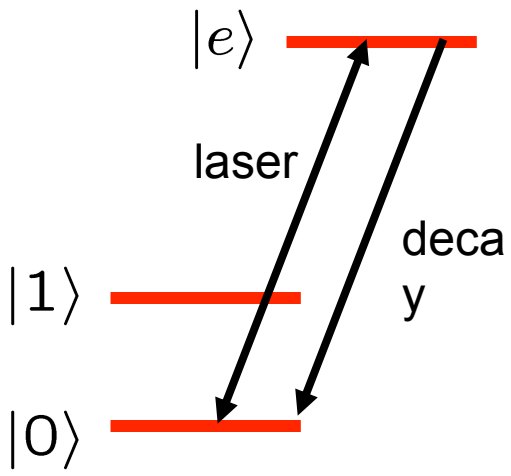


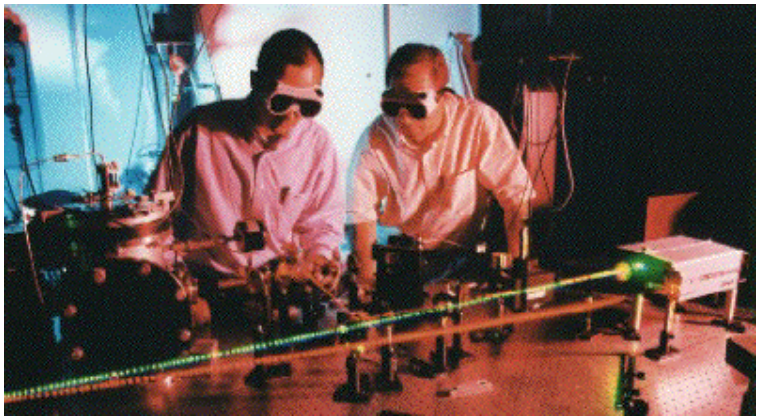
. Well defined qubits in a scalable architecture

Solid state qubits seem to have a huge advantage for scalability.

Measuremen





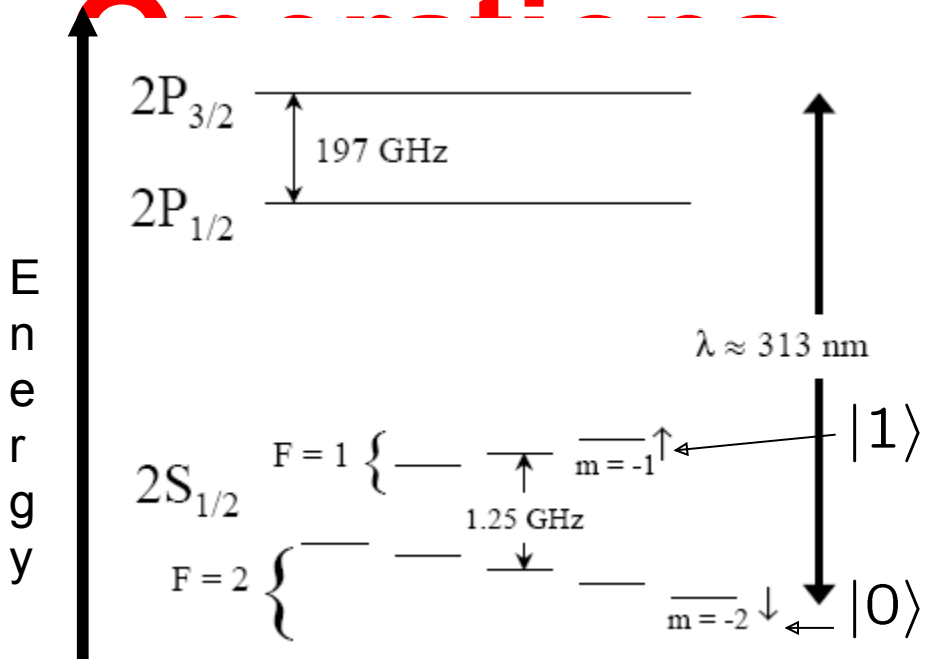


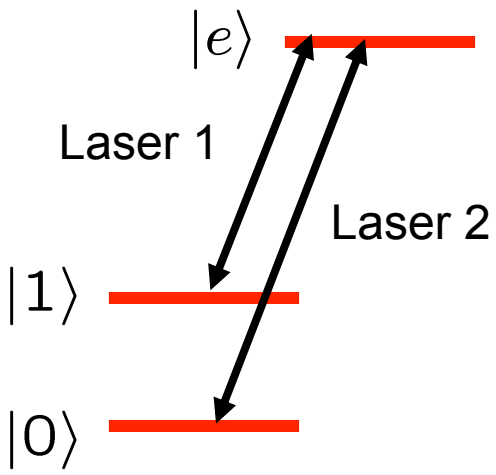
Detecting florescence implies in state 0

99.99% efficiency

5. Have the ability to perform high quality measurements ♠

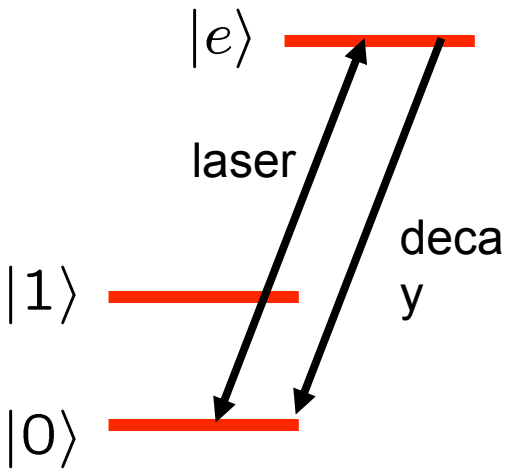
Single Qubit Operations



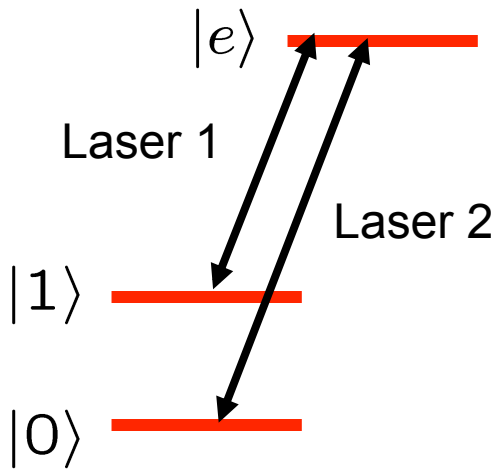


Allows any one qubit unitary operations $\begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}$

Initialization



measure

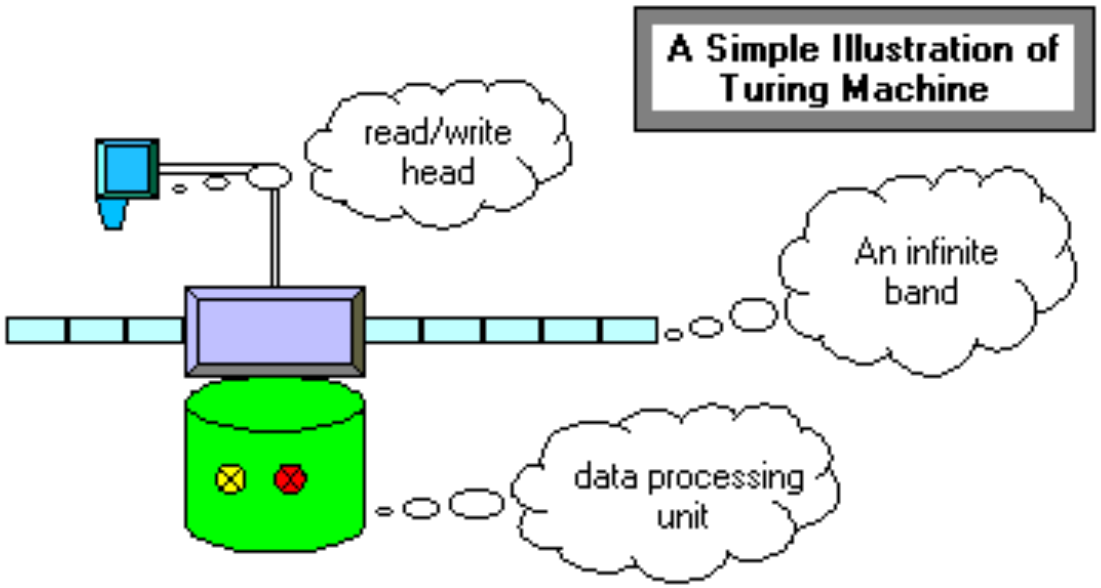


If not in zero state, flip

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

2. The ability to initial the system to a deterministic state. ♠

Universal C



1. Turing machine reads state of tape at current position.
2. Based on this reading and state of machine, Turing machine writes new symbol at current position and possibly moves left or right.

Certain Turing machines can perform certain tasks.

A Universal Turing Machine can act like any other possible Turing machine (i.e. it is programmable)

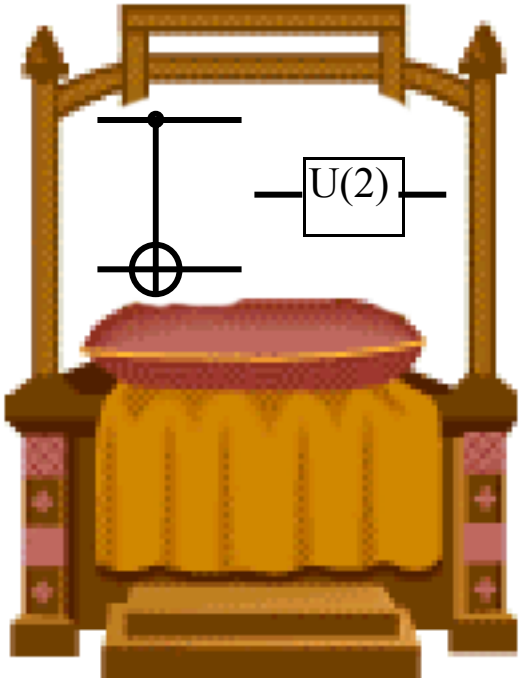
Universal Quantum Computer

Universal Quantum Computer

•a quantum computer which can be programmed to perform any algorithmic manipulation on quantum information.

Set of Universal Quantum Gates

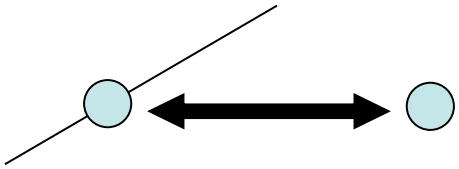
•a set of operations/gates which, acting on the quantum information, can be used to implement (to any desired accuracy) any unitary evolution of the quantum info.



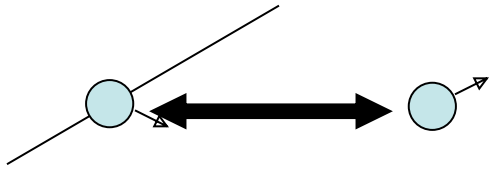
The Royal King and Queen of Universal Quantum Gates

CNOT and 1-qubit rotations

Coupling Two Qubits

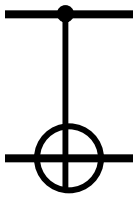


stationary



sloshing mode

These modes can be used as a bus between the qubits.



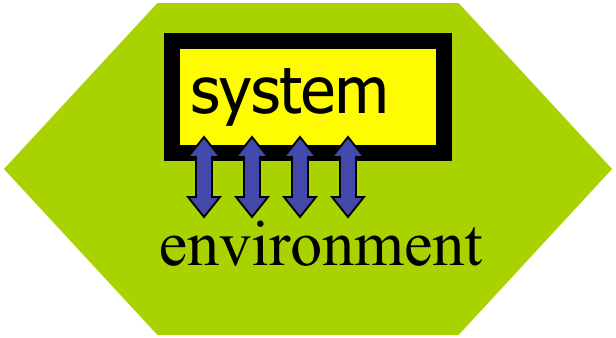
$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

4. Have the ability to perform a universal set of quantum gates ♠

What is the Problem?

3. Have faster control over the system than error processes in the system.

Real quantum systems are open quantum systems!



Quantum systems readily couple to an environment…

System decoheres:

$|0\rangle$
$|1\rangle$
qubits

$\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$

Quantum

0
1
bits

50% **0** 50% **1**

Classical

The Decoherence Problem (1996)

The Problem

Decoherence is a lot like classical noise, BUT:

Strong coupling to control devices needed to enact computations



Strong coupling to environment causes decoherence

Yingyang of quantum computing

Quantum Computing is Bunk

Ways Quantum Computers Fail to Quantum Compute

Decoherence

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\text{environment}\rangle \rightarrow \alpha|0\rangle \otimes |\text{environment}_A\rangle + \beta|1\rangle \otimes |\text{environment}_B\rangle$$

Lack of Unitary Control

attempting to apply unitary evolution **U** instead results in **V**

or (worse) results in non-unitary evolution

$$\boldsymbol{\rho} \rightarrow \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger$$

Measurements are faulty

measurement result is noisy, incorrect result obtained

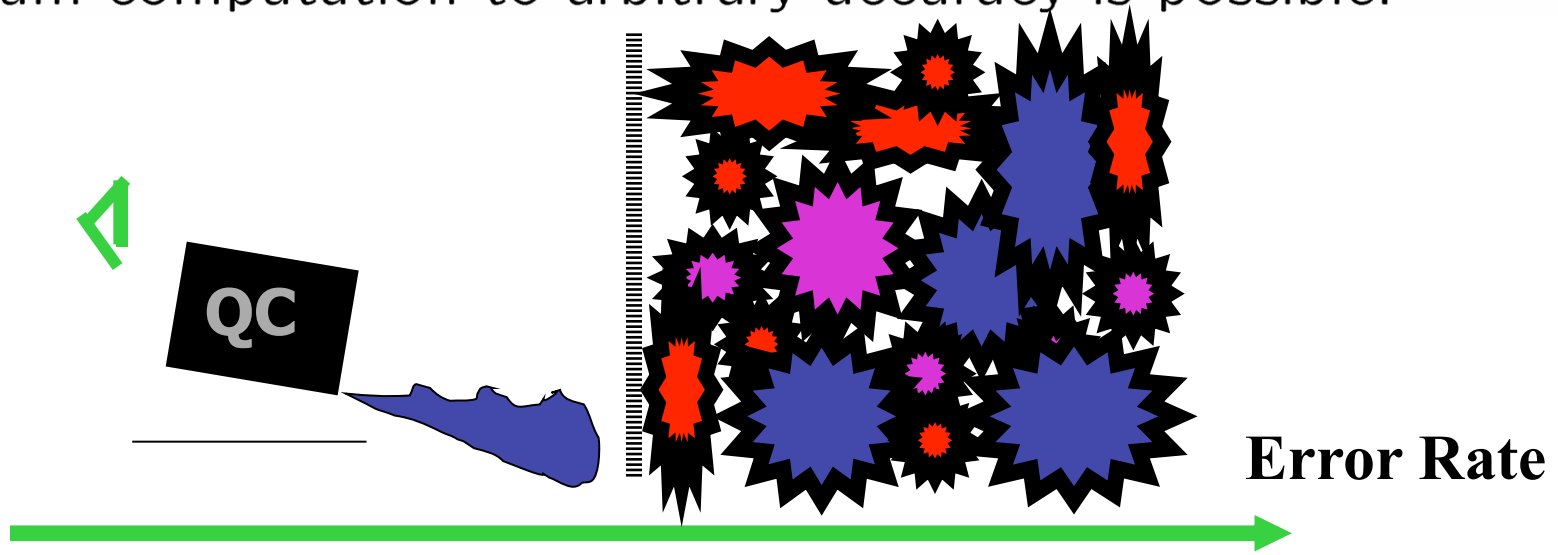
Quantum Computing Disappearing Act

qubits disappear (leakage of computing states)

The Quantum Solution (1995-96)

Threshold

If the error rate (typical number of operations per decoherence or failure event) for single qubit errors is smaller than ϵ, then quantum computation to arbitrary accuracy is possible.



Ion Trap Parameters

Decoherence rate for qubits: 1 minutes
Gate speed: 10 microseconds
Decoherence rate for bus: 100 microseconds to 100 milliseconds
Measurement errors: 0.01%

3. Have faster control over the system than error processes in the system. ♠

State of the Art



letters to nature

Experimental entanglement of four particles

C. A. Sackett*, D. Kielpinski*, B. E. King*†, C. Langer*, V. Meyer*, C. J. Myatt*‡, M. Rowe*, Q. A. Turchette*‡, W. M. Itano*, D. J. Wineland* & C. Monroe*

* Time and Frequency Division, National Institute of Standards and Technology, Boulder, Colorado 80303, USA
† Atomic Physics Division, NIST, Gaithersburg, Maryland 20899, USA
‡ Research Electro-Optics, 1855 S. 57th Ct., Boulder, Colorado 80301, USA

NIST Boulder

A Critical Ghost



All papers on quantum computing should carry a footnote: *"This proposal, like all proposals for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work."*

Rolf Landauer
IBM

- Maintenance of giganto-coherence?
- Faulty quantum gates?
- Do we understand the physics of quantum errors in the system?

Nature abhors a quantum computer?



Analog Computers

Compute by adding, multiplying real infinite precision numbers.

This can be used to solve NP complete problems in polynomial time!

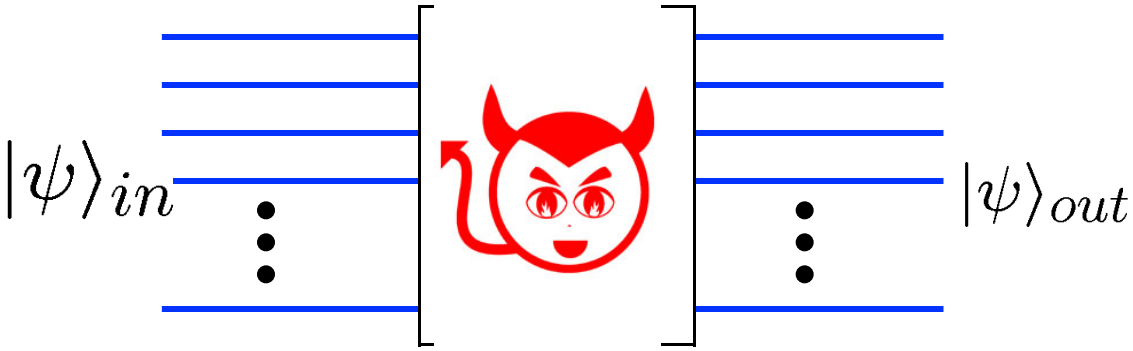
This, however is NOT a realistic model of computation.

Why? Infinite precision is requires, as far as we know, infinite resources! Noise destroys the speedup.

Is quantum computing an analog computer?

The resolution of this is the subject of quantum error correction.

Don't Eat That Apple



plus: simple
minus: unrealistic
plus: essential ideas

Lucifer's channel:

set of unitaries: $\mathcal{S} = \{\mathbf{U}_1 = \mathbf{I}, \mathbf{U}_2, \mathbf{U}_3, \ldots, \mathbf{U}_k\}$

probabilities: $p_1, p_2, p_3, \ldots, p_k$, $p_i \geq 0$, $\sum_i p_i = 1$

devil applies unitary $\mathbf{U}_i$ with probability p_i

no which $\mathbf{U}_i$ information available

Identity

simplest example (single qubit):

$$\mathcal{S} = \{\mathbf{I}, \mathbf{X}, \mathbf{Z}\},\ p_i > 0$$

$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

How to preserve $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?

less silly? example (two qubits):

$$\mathcal{S} = \{\mathbf{I}, \mathbf{Z}_1\mathbf{Z}_2\},\ p_i > 0$$

$$\mathbf{Z}_1\mathbf{Z}_2 = \begin{array}{c} \begin{matrix} |00\rangle & |01\rangle & |10\rangle & |11\rangle \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

notation

$$\mathbf{P}_i = \mathbf{I} \otimes \cdots \otimes \mathbf{I} \otimes \underbrace{\mathbf{P}}_{ith\ qubit} \otimes \mathbf{I} \otimes \cdots \otimes \mathbf{I}$$

Encode information into first qubit? $(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle$

$$\mathbf{Z}_1\mathbf{Z}_2(\alpha|0\rangle + \beta|1\rangle)|0\rangle = (\alpha|0\rangle - \beta|1\rangle)|0\rangle$$

Solution: encode into subspace $|00\rangle$, $|11\rangle$

$$\mathbf{Z}_1\mathbf{Z}_2(\alpha|00\rangle + \beta|11\rangle) = \alpha|00\rangle + \beta|11\rangle$$

Qubit can be protected by encoding into a subspace

The Story of the Ghost



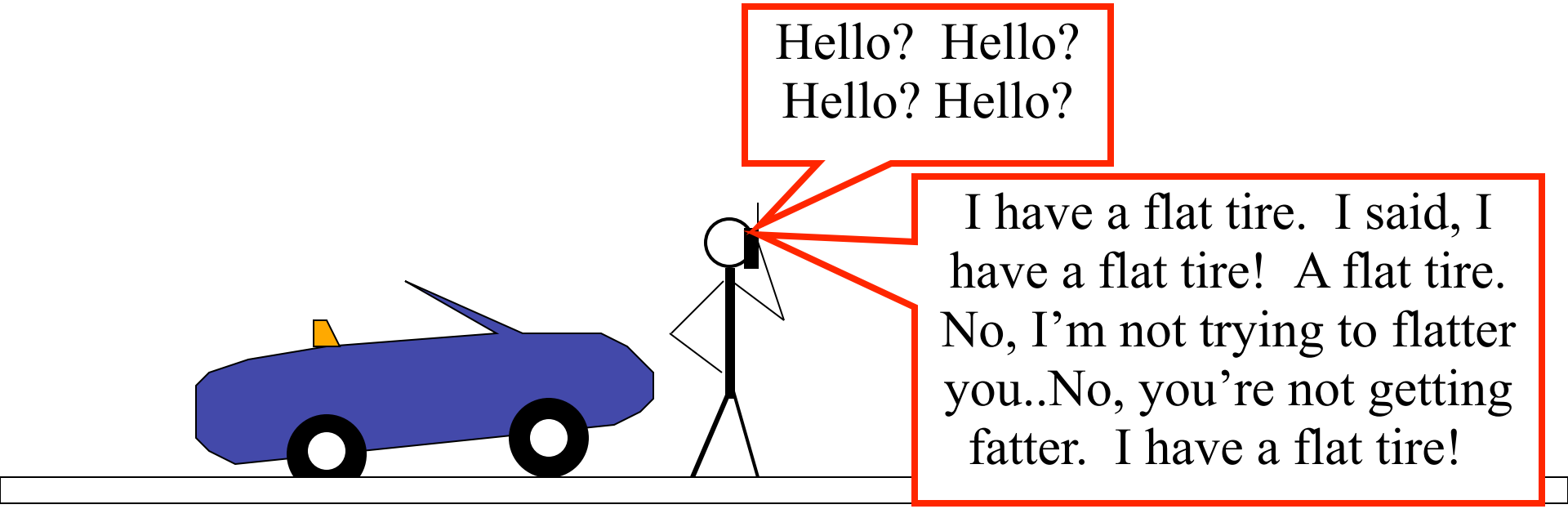


You are protecting your quantum information against a crazy noise model! Z1Z2? If this is all nature can throw at you, then pigs can fly.

Rolf Landauer
IBM

Can we use encoding to protect quantum information from more realistic noise models?

Noisy Cell Phone



Communication over a noisy CHANNEL can be overcome via

ENCODING

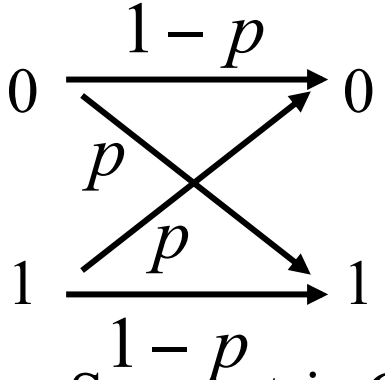
"Hello?" = "Hello? Hello? Hello? Hello?" [using redundancy to encode "Hello"]

Simple Repetition Code

Encode: $0 \rightarrow 000\ldots000$

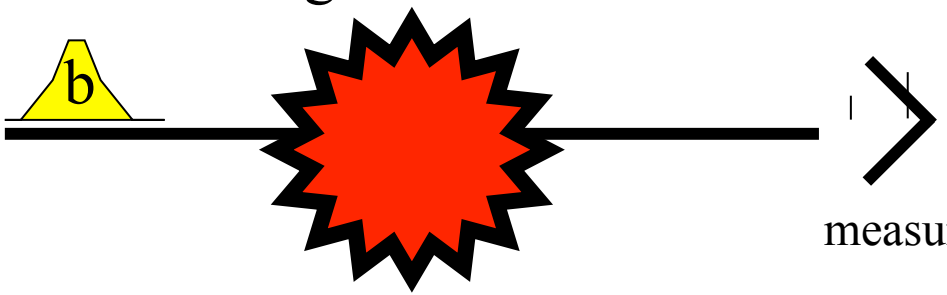
$1 \rightarrow 111\ldots111$

n copies



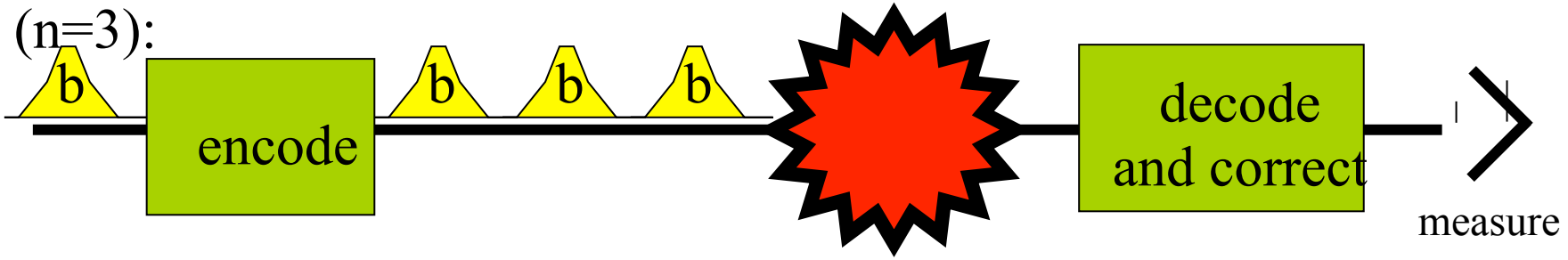
Binary Symmetric Channel

No encoding:



Probability of error = p

Encoding (n=3):



Probability of error

$$= 3p^2(1-p) + p^3 = 3p^2 - 2p^3 < p, p < \tfrac{1}{2}$$

1994 Reasons to be a Pessimist

No cloning:



Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

No quantum repetition code:

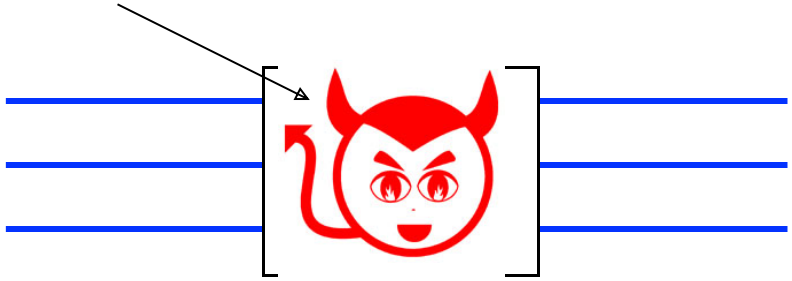
$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

Measurement destroys coherence:

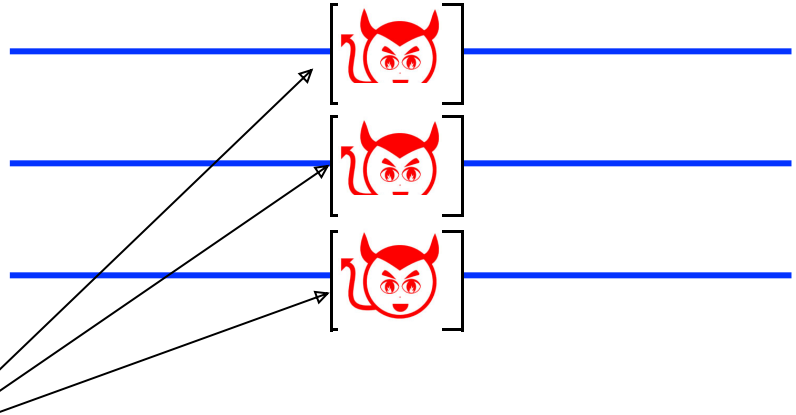
How can one decode without destroying the information?

Unrealistic Realistic Channel

$\mathcal{S} = \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3\},\ p_1 = 1 - 3\epsilon, p_2 = p_3 = p_4 = \epsilon$



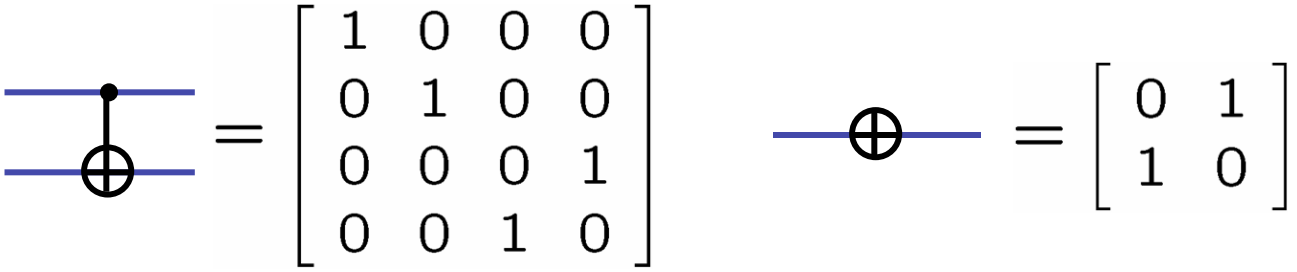
≀



$\mathcal{S} = \{\mathbf{I}, \mathbf{X}\},\ p_1 = 1 - \epsilon, p_2 = \epsilon$

Baby Steps

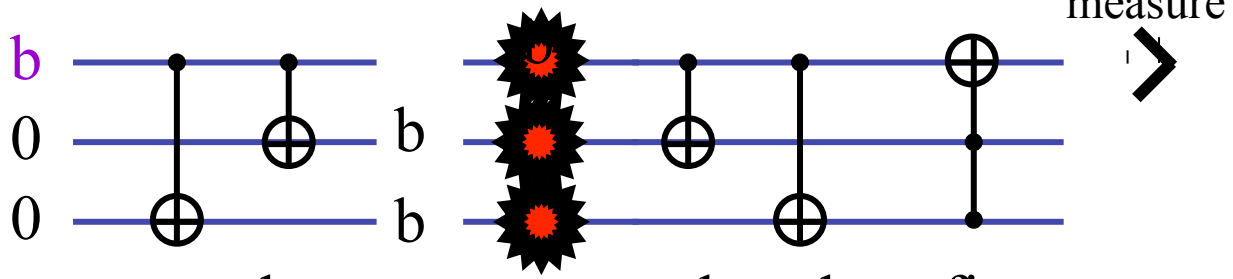




$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

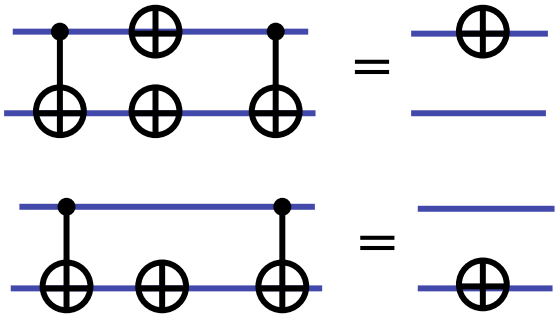
WWCCD? (What Would Classical Coders Do?)

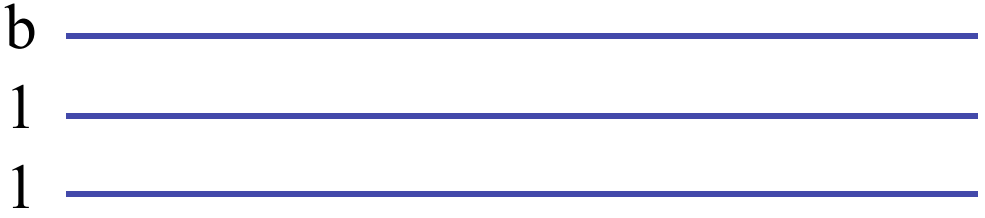
0 □000, 1 □ 111



100 □ 111□101 □ 110 □ 110

identities



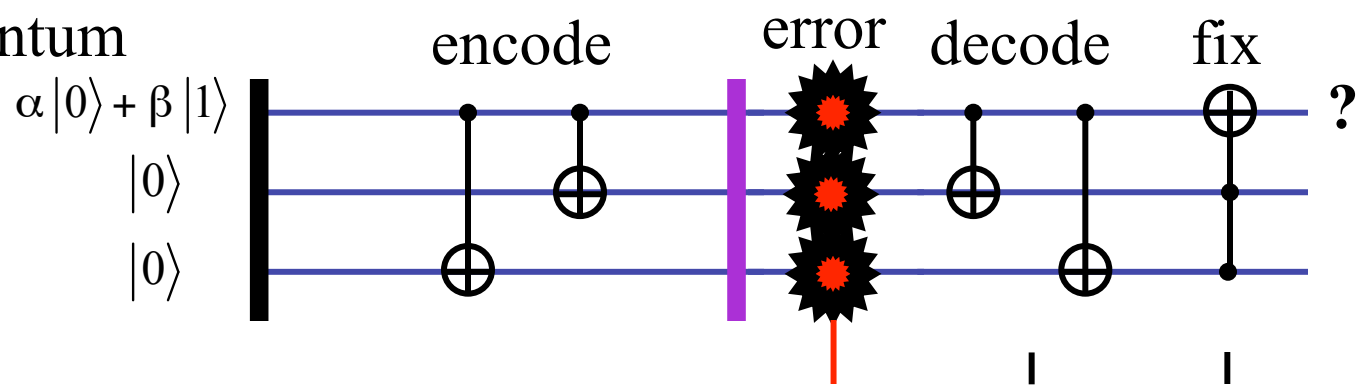


error #@%

Naïve

4. operator identities still hold

Lets be naïve, take classical and move over to quantum



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

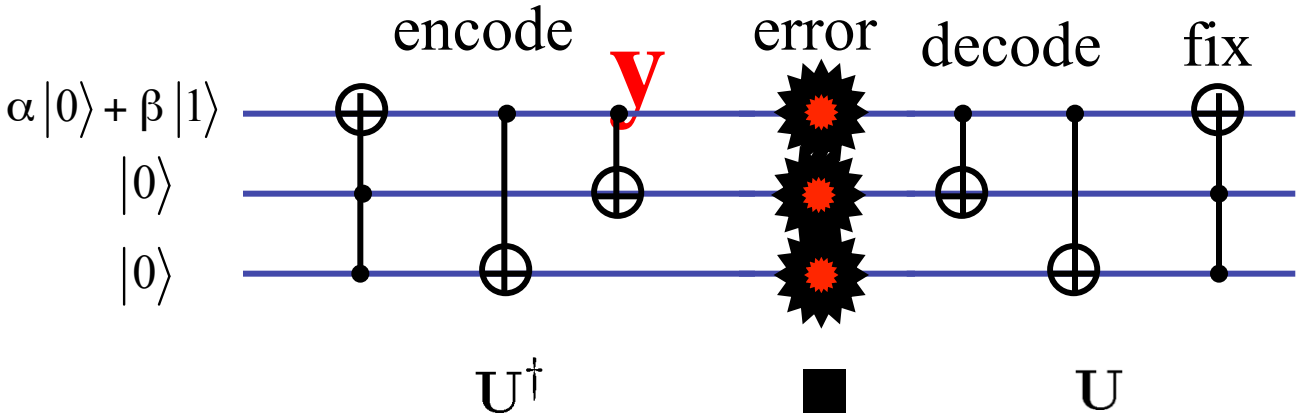
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

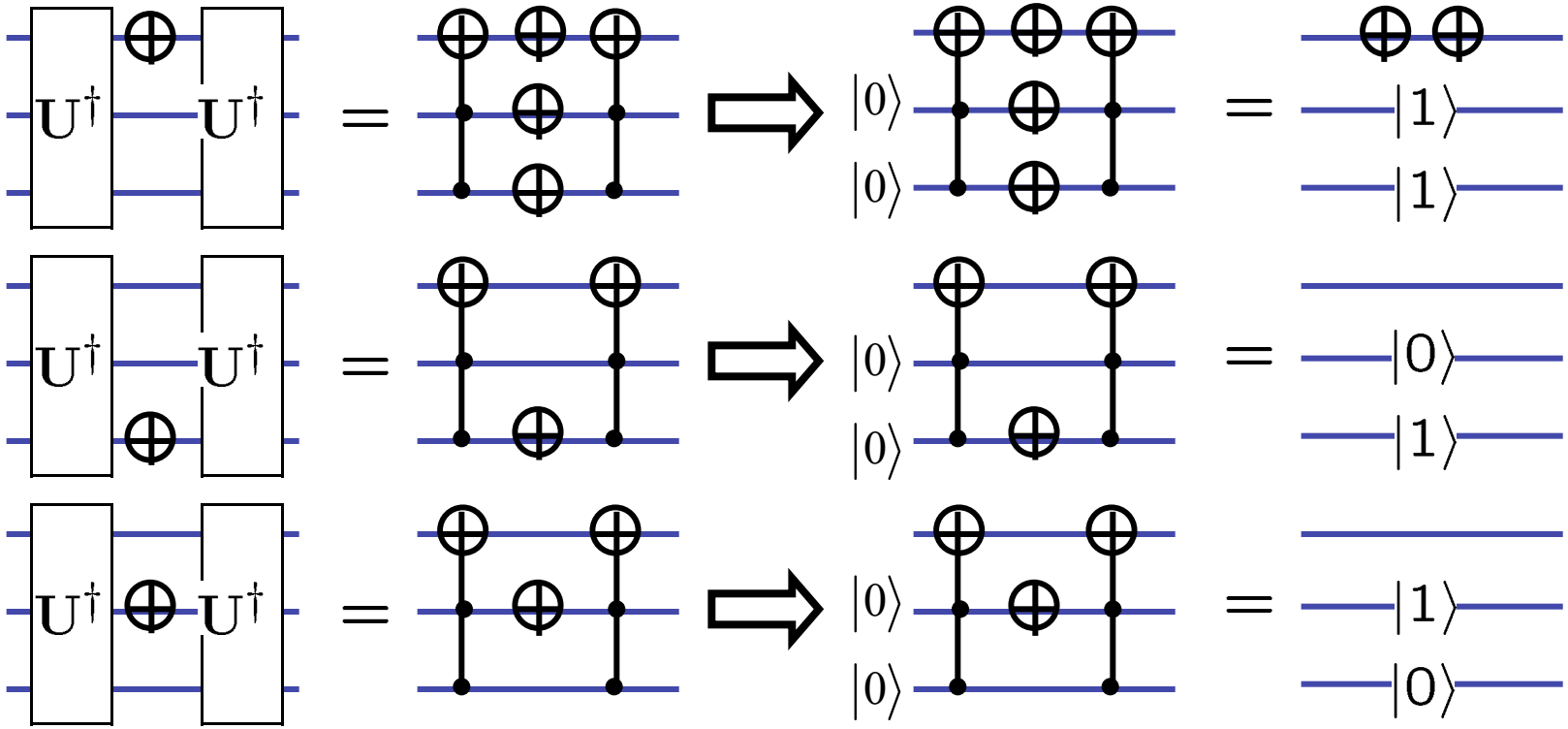
3. syndrome

$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

Identit



But this is just a basis change: $UEU^\dagger$, E is the error



OK Wise Guy

What about "phase" errors? $\boldsymbol{\rho}_{out} = (1-p)\boldsymbol{\rho}_{in} + p\boldsymbol{\sigma}_z\boldsymbol{\rho}_{in}\boldsymbol{\sigma}_z$

phase error: $\boldsymbol{\sigma}_z(\alpha|0\rangle + \beta|1\rangle) = (\alpha|0\rangle - \beta|1\rangle)$ …sort of not classical error

$$\mathcal{S} = \{\mathrm{I}, \mathrm{X}_1, \mathrm{X}_2, \mathrm{X}_3\},\ p_1 = 1-3\epsilon, p_2 = p_3 = p_4 = \epsilon$$

Wise guy says "basis change please":

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

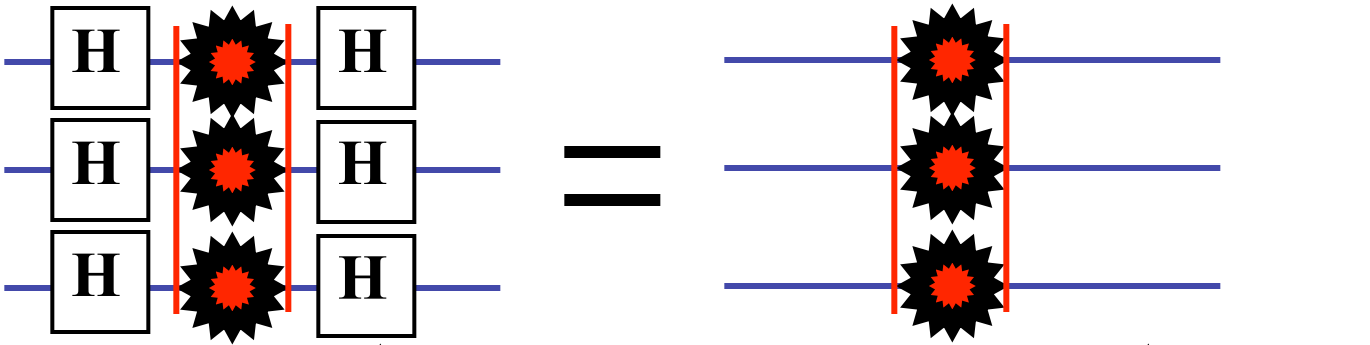
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\boldsymbol{\sigma}_z|+\rangle = |-\rangle$$

$$\boldsymbol{\sigma}_z|-\rangle = |+\rangle$$

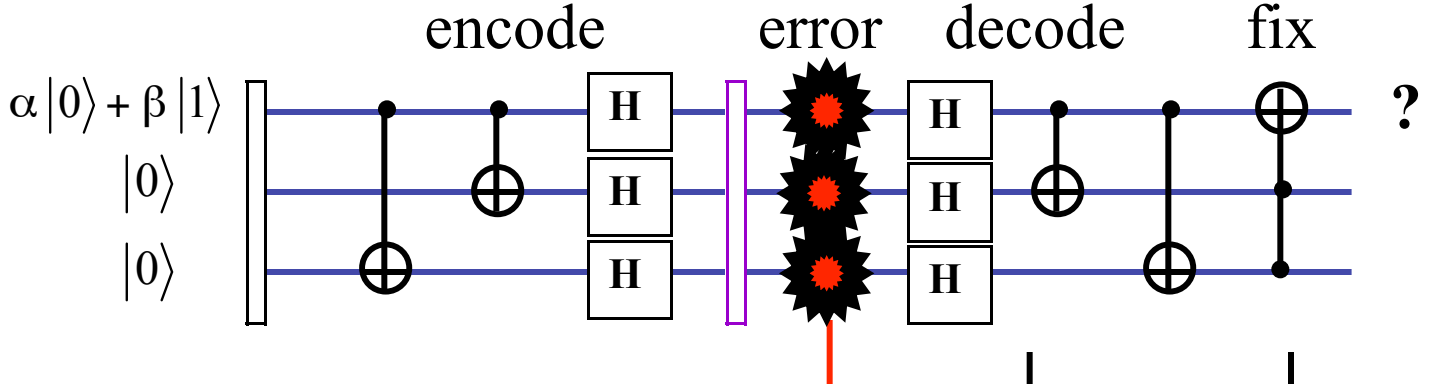
looks like bit flit error in this new basis!

$$\mathrm{H}\boldsymbol{\sigma}_z\mathrm{H} = \boldsymbol{\sigma}_x, \quad \mathrm{H} = |+\rangle\langle 0| + |-\rangle\langle 1|$$



phase errors $= \boldsymbol{\sigma}_z$ bit flip errors $= \boldsymbol{\sigma}_x$

Molly: "I love you, I really love you"
Sam: "Ditto."



$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{ZII}{\rightarrow} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IZI}{\rightarrow} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IIZ}{\rightarrow} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

encoded into subspace: $|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

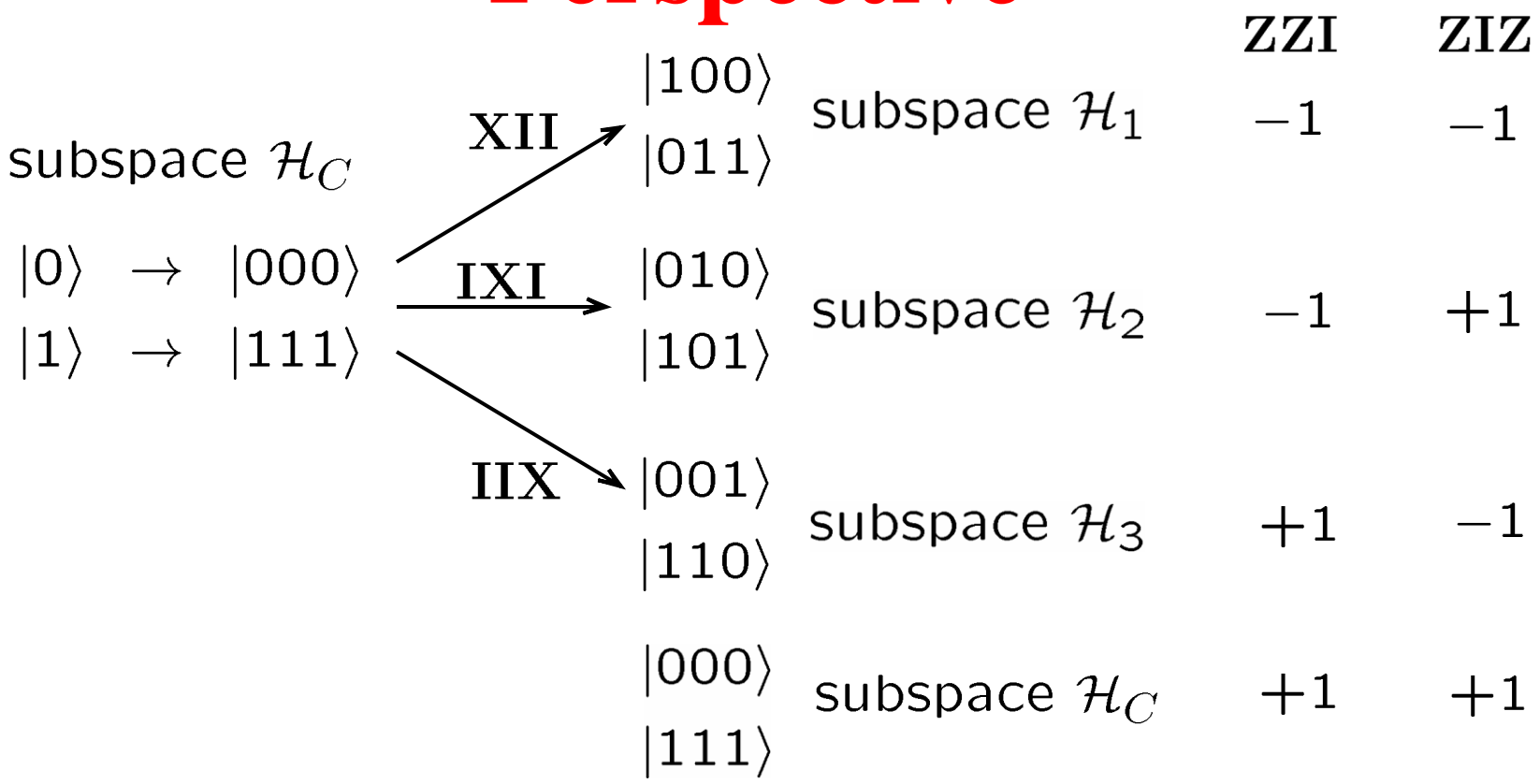
(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. syndrome

Perspective



subspaces $\mathcal{H}_C$, $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ are orthogonal

$\dim \mathcal{H}_C = \dim \mathcal{H}_1 = \dim \mathcal{H}_2 = \dim \mathcal{H}_3 = 2$

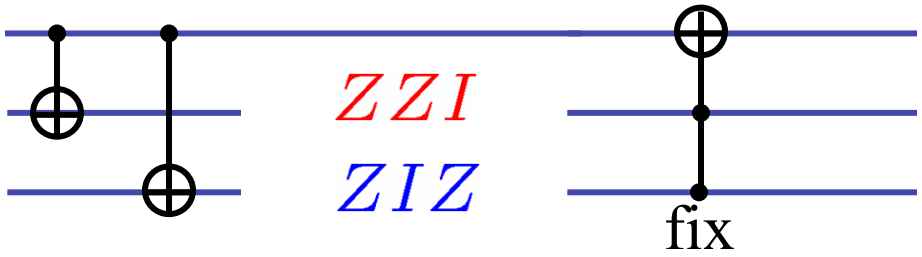
Orthogonal subspaces can be distinguished by measurements

without measuring information encoded into the subspace!!!

Not An Optical Illusion



		b	ZZ
$\|00\rangle$	$\rightarrow$	$\|00\rangle$	$+1$
$\|01\rangle$	$\rightarrow$	$\|01\rangle$	-1
$\|10\rangle$	$\rightarrow$	$\|11\rangle$	-1
$\|11\rangle$	$\rightarrow$	$\|10\rangle$	$+1$



		XII	$\|100\rangle$	$\|111\rangle$	$\|011\rangle$
			$\|011\rangle$	$\|011\rangle$	$\|111\rangle$
$\|0\rangle$	$\rightarrow$ $\|000\rangle$	IXI	$\|010\rangle$	$\|010\rangle$	$\|010\rangle$
$\|1\rangle$	$\rightarrow$ $\|111\rangle$		$\|101\rangle$	$\|110\rangle$	$\|110\rangle$
		IIX	$\|001\rangle$	$\|001\rangle$	$\|001\rangle$
		error	$\|110\rangle$	$\|101\rangle$	$\|101\rangle$

Encoding Away Your Ills

3 qubit bit flip code	3 qubit phase flip code
$\|0\rangle \rightarrow \|000\rangle$	$\|0\rangle \rightarrow \|+++\rangle$
$\|1\rangle \rightarrow \|111\rangle$	$\|1\rangle \rightarrow \|---\rangle$

phase errors **ZII**, **IZI**, **IIZ** act as ■ on bit flip code qubits:

$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

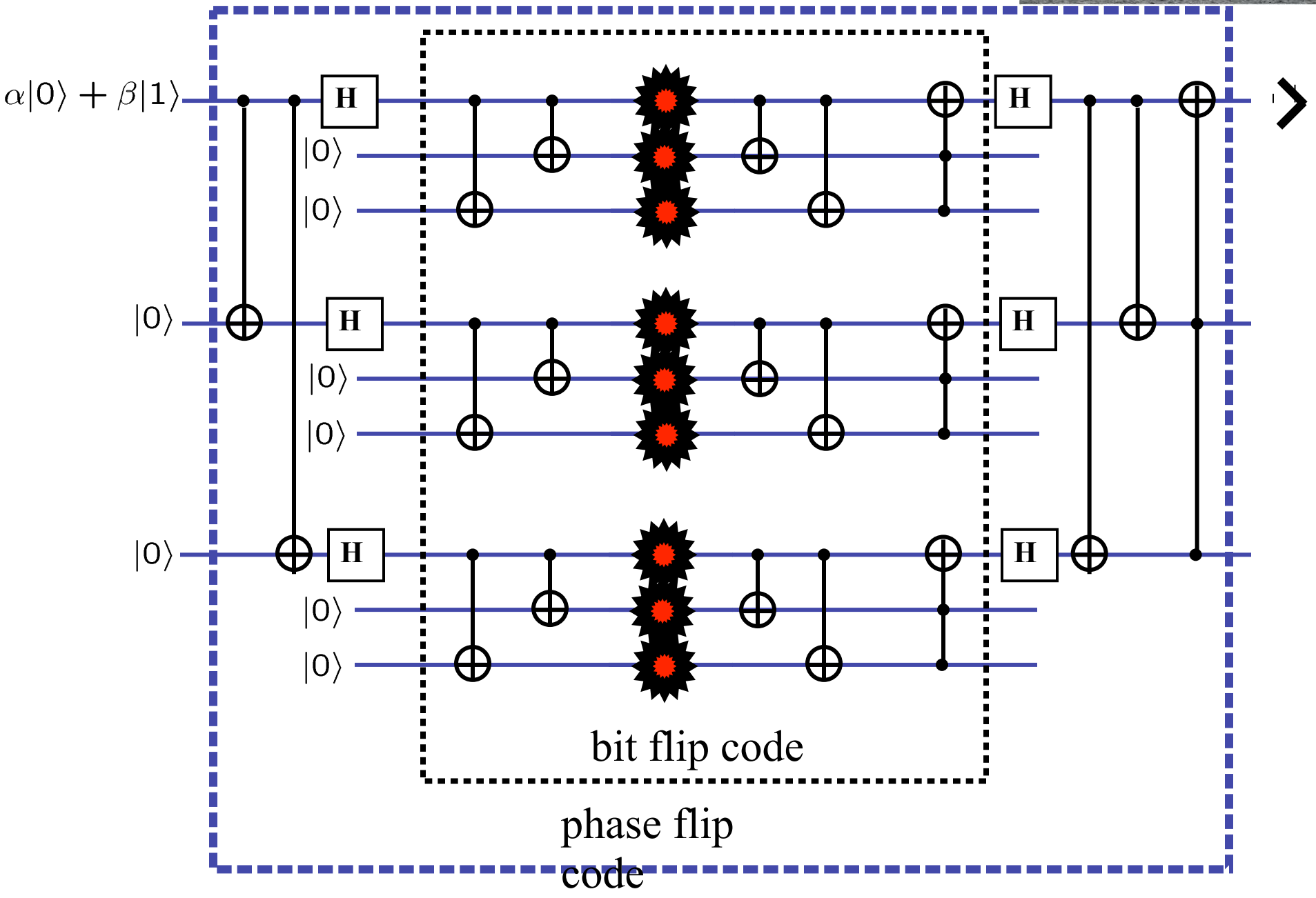
define:
$$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor Code: (Peter Shor, 1995)

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

Inside Shor




$\alpha|0\rangle + \beta|1\rangle$
$|0\rangle$
$|0\rangle$
H
H
H
H
H
H
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
bit flip code
phase flip
code

Linearity of Errors

We have only discussed two types of errors, bit flips and phase flips. What about "general" errors?

Theorem of digitizing quantum errors:

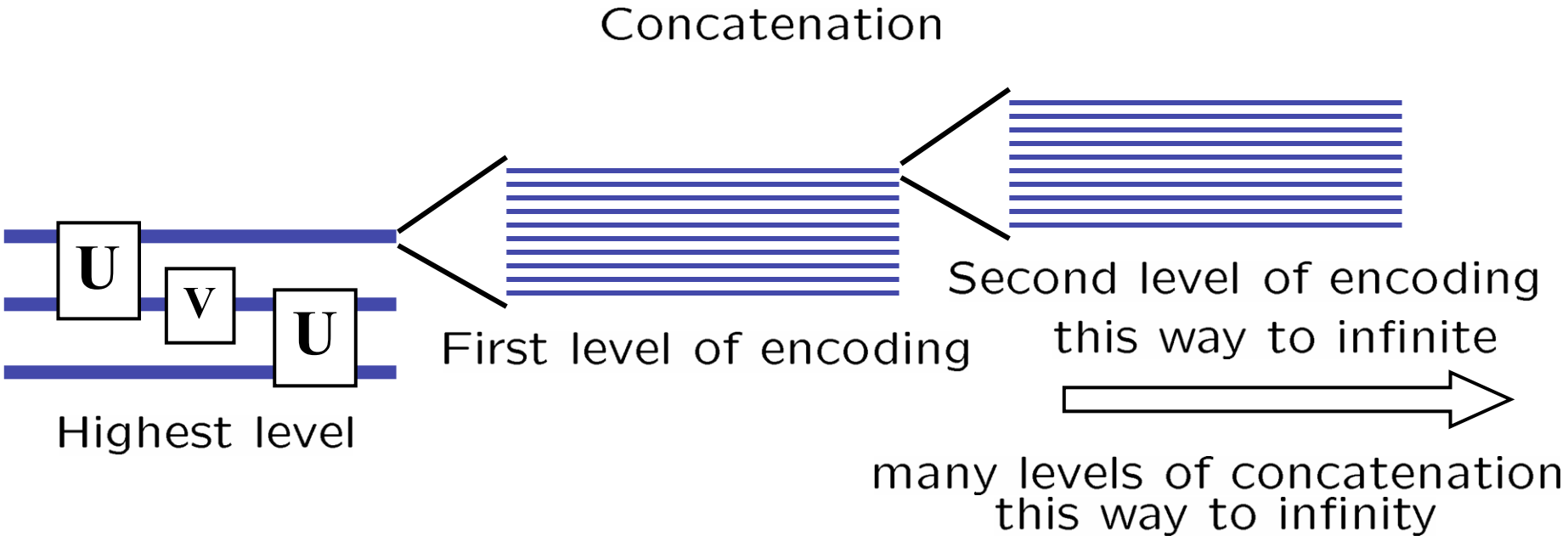
If we can correct errors in some set, $\mathcal{S}$ then we can correct any linear complex combination of such errors.

$$\text{set of unitaries: } \mathcal{S} = \{\mathbf{U}_1 = \mathbf{I}, \mathbf{U}_2, \mathbf{U}_3, \ldots, \mathbf{U}_k\}$$

While errors may form a continuous set, we only need to correct a discrete set of these errors

Perfection Through Concatenation

Idea 2: Error correct these encoded qubits to fix 'errors'



Threshold Theorem for Quantum Memory

If the single qubit error rate is below some threshold p_t, then we can store quantum states to accuracy ϵ using poly$(\log(\epsilon^{-1}))$ qubits.

Quantum Error Correction

The insight that quantum computers could be defined in the presence of noise (the full theory is called fault-tolerant quantum computation) is why we have been justified in using the quantum circuit model.

Quantum error correction justifies calling a quantum computer a digital computer.

Whence Physics?

RANT MODE ON

Today: similar situation to early days of classical computation (threshold theorems but no physics!)

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

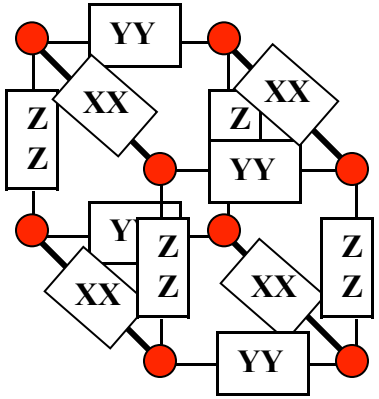
not all physical systems are equally good for computation

there exists systems whose <u>PHYSICS</u> guarantees their ability to enact robust classical computation.

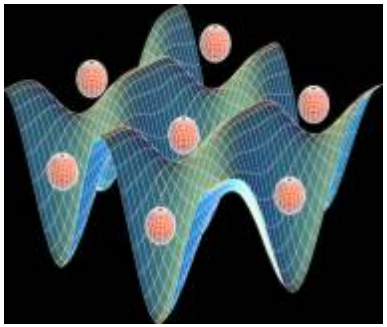
THE BILLION DOLLAR QUANTUM QUESTION:

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

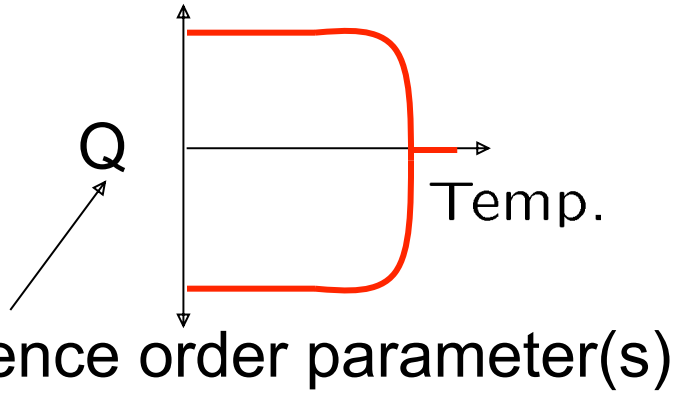
Self-Correcting Quantum Computers



Quantum many-body systems which have excitations which are string-like and are self-correct, but into which we can encode quantum information?



optical lattice



[Bacon, Ph.D. thesis, U.C. Berkeley 2001]
[Bacon, "Quantum Error Correcting Subsystems" in preparation]

Quantum Cryptography

We saw that quantum computers defeat many public key cryptosystems. Luckily quantum theory also provides an alternative, known as quantum cryptography.

Goal: a manner in which Alice and Bob can share secret key such that they can detect if an eavesdropper can be detected.

Quantum Cryptography





Alice generates 2n bits with equal probability

$$b_1, b_2, \ldots, b_n$$
$$x_1, x_2, \ldots, x_n$$

The first of these bits labels a basis choice and the second labels a wave function choice. Alice prepares n qubits:

b_i	x_i	Alice's qubit
0	0	$\|0\rangle$
0	1	$\|1\rangle$
1	0	$\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$
1	1	$\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$

Quantum Cryptography





Alice sends her n qubits to Bob.

Alice then announces via a public channel what basis she measured in: the b bitstring.

If Bob measures his qubits in the same basis, he will end up with results which exactly match Alice's bit string

$$x_1, x_2, \ldots, x_n$$

They can then reveal a few of their bits at random to check whether someone has been eavesdropping.

If not eavesdropping, the rest of their bits are a shared

Quantum Cryptography





Eve sees a procession of qubits in the computational or plus/minus basis. Eve does not know the basis.

Intuition: If Eve tries to measure this qubit, since she doesn't know what basis to measure in, sometimes she will make measurement in the wrong basis and this can be detected by Alice and Bob.

Quantum Cryptography





0	1	1	0	0
1	0	0	0	1

$|1\rangle \quad |+\rangle \quad |+\rangle \quad |0\rangle \quad |1\rangle$

b_i	x_i	Alice's qubit
0	0	$\|0\rangle$
0	1	$\|1\rangle$
1	0	$\|+\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$
1	1	$\|-\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$

0	0	1	1	1

Eve's basis

$|1\rangle$ 50% $|0\rangle$ 50% $|1\rangle$ $\quad |+\rangle$ $|+\rangle$ $|+\rangle$ 50% $|+\rangle$ 50% $|-\rangle$

State after Eve's measurement

Quantum Cryptography





Eve sees a procession of qubits in the computational or plus/minus basis. Eve does not know the basis.

Proof of security, with certain key generation rate, against all types of Eve's attacks.

Quantum Cryptography



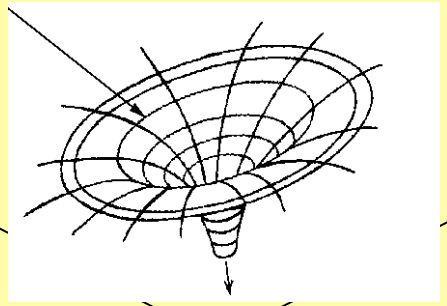




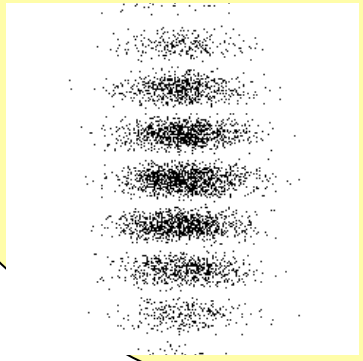
Black Holes Information Paradox

Three Revolutions of Fundamental Modern Physics

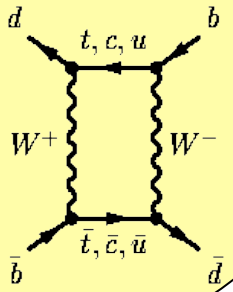
Relativity
Special and General



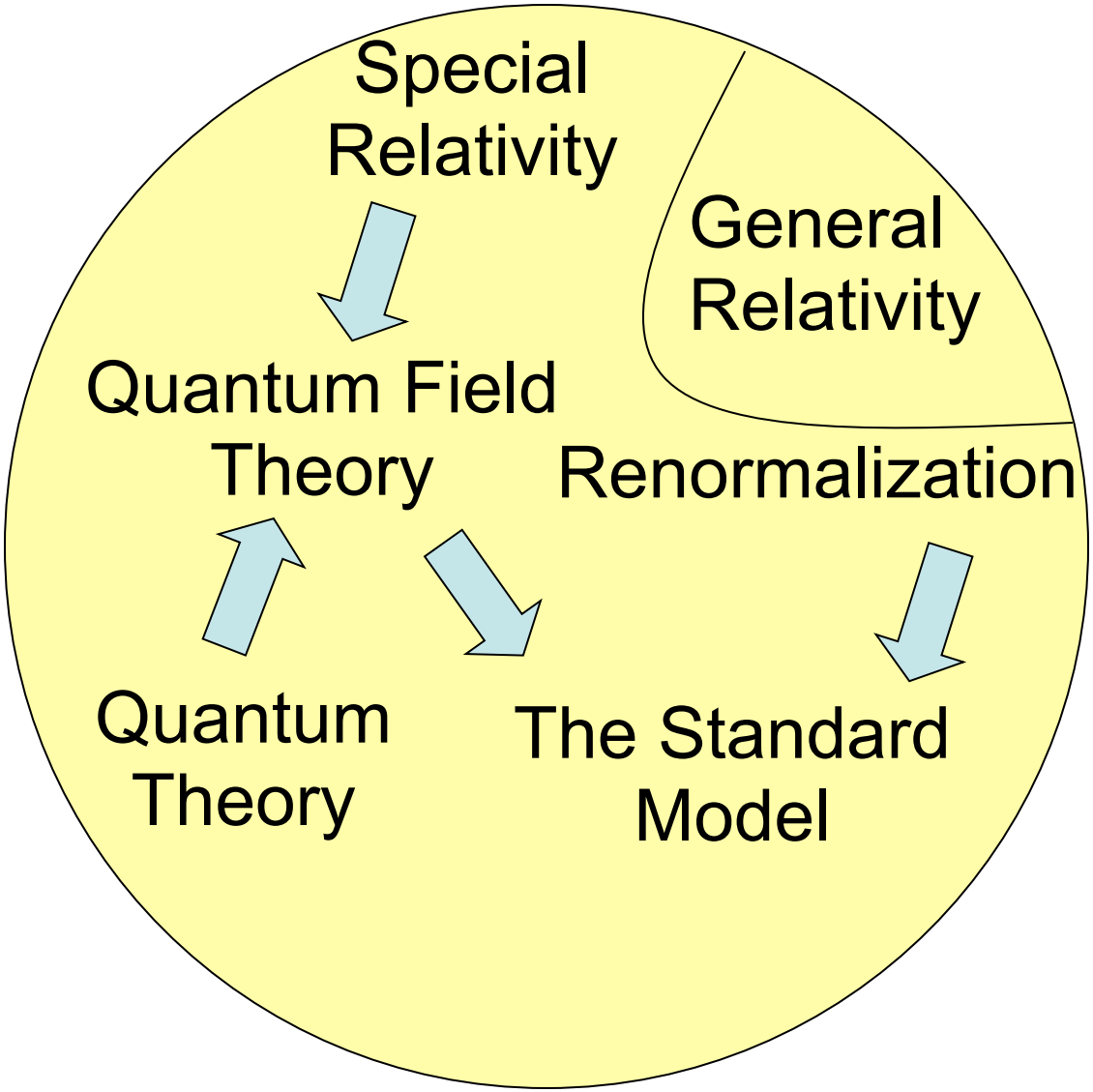
Quantum
Theory

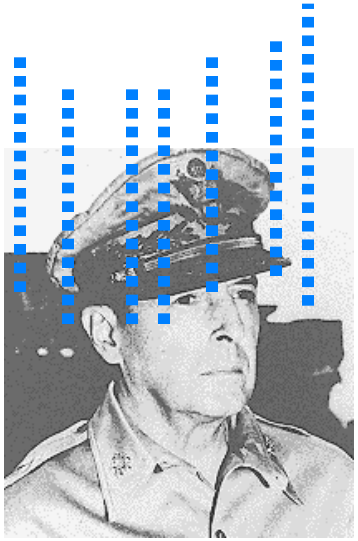


The Standard
Model



Three Revolutions of Fundamental Modern Physics



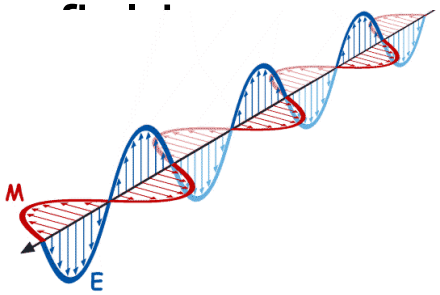


The Physics

Quantum Field Theory

dynamic variables

particle

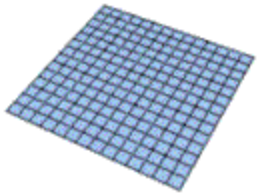


$$\phi(x)\ \psi_\mu(x)\ A_\mu(x)\ G^\alpha_\mu(x)$$

defined over some space-time

General Relativity

metri



$$g\mu,\nu$$

space-time itself!

Black Holes

Blackholes: $ds^2 = \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} c^2 dt^2$

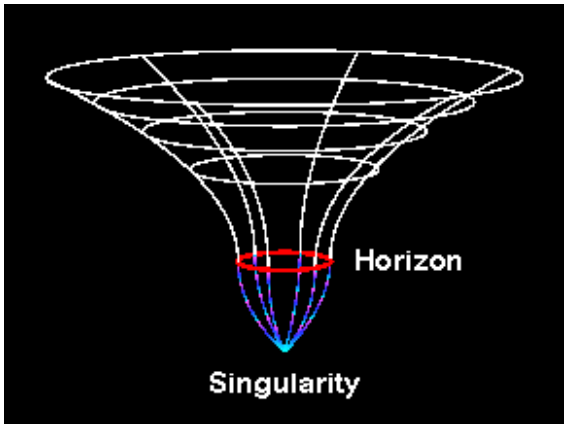
If we cram mass inside $r < r_s = \frac{2GM}{c^2}$
we create a blackhole.

Two regions:

A. outside of the black hole.

B. Inside the horizon of the black hole.

Things can go from A to B, but not from B to A



At the center of the black hole, the general relativity solution becomes singular. This is scary and no one knows what to do about this.

Black Holes Have no Hair?

Blackholes: $ds^2 = \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} c^2 dt^2$

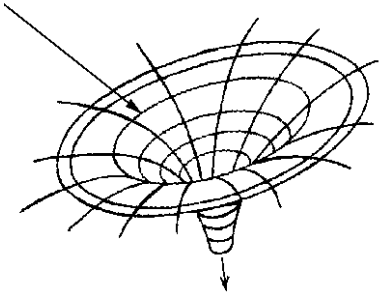
If we cram mass inside $r < r_s = \frac{2GM}{c^2}$
we create a blackhole

Classically, black holes have only three properties which are accessible to an observer outside of the black hole:
Mass M, Charge Q, Angular momentum L

We say that a “black hole has no hair.”

All other information about how we formed the black hole has disappeared except these three numbers.

Black Holes Thermodynamics



Throwing stuff into a black hole will increase it's mass

This will increase the radius of the black hole

$$r_s = \frac{2GM}{c^2}$$

Second law of thermodynamics: the entropy of a closed system can only increase.

Entropy measures roughly the "degrees of freedom" of a physical system.

Entropy of a black hole of area A: $S = \frac{Ak_B c^3}{4\hbar G}$

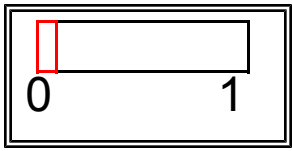
k_B Boltzman's constant

G Newton's constant

$\hbar$ Planck's constant

c speed of light

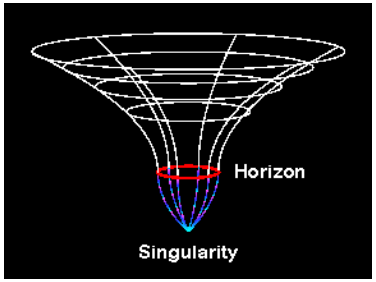
Planck Length



General Relativity

Blackholes: $ds^2 = \left(1 - \frac{2GM}{rc^2}\right)^{-1} dr^2 + r^2 d\Omega^2 - \left(1 - \frac{2GM}{rc^2}\right)^{-1} c^2 dt^2$

If we cram mass inside $r < r_s = \frac{2GM}{c^2}$
we create a blackhole



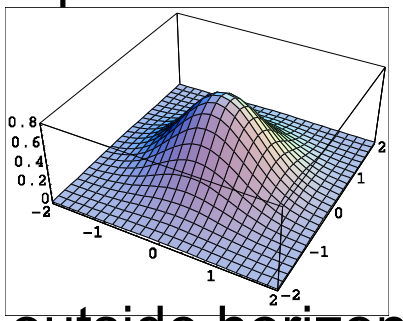
Any mass blackhole possible

Quantum Field Theory

$\Delta x \Delta p \geq \frac{\hbar}{2}$ localize to diameter d → large momentum possible

large momentum mc → particle creation

Compton length: $l_c = \frac{\hbar}{Mc}$



Black holes of small mass such that Compton length is outside horizon?

$M_P = \sqrt{\frac{\hbar c}{G}} \approx 10^{-8}$ kg

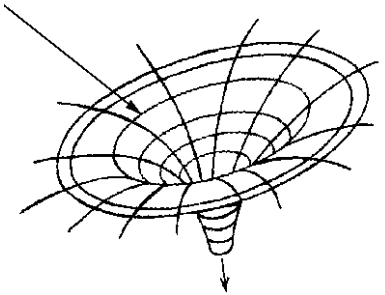
Planck Mass

$L_P = \sqrt{\frac{G\hbar}{c^3}} \approx 10^{-35}$ m

Planck Length

Black Holes Thermodynamics

Entropy of a black hole of area A: $S = \frac{Ak_B c^3}{4\hbar G}$



k_B Boltzman's constant

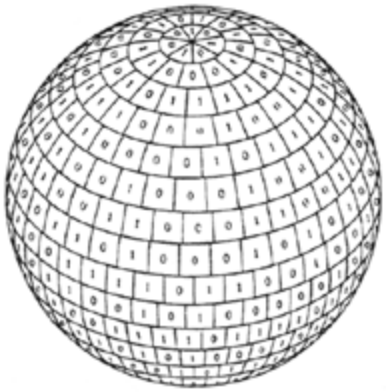
G Newton's constant

$\hbar$ Planck's constant

c speed of light

$$L_P = \sqrt{\frac{G\hbar}{c^3}} \approx 10^{-35} \text{ m}$$

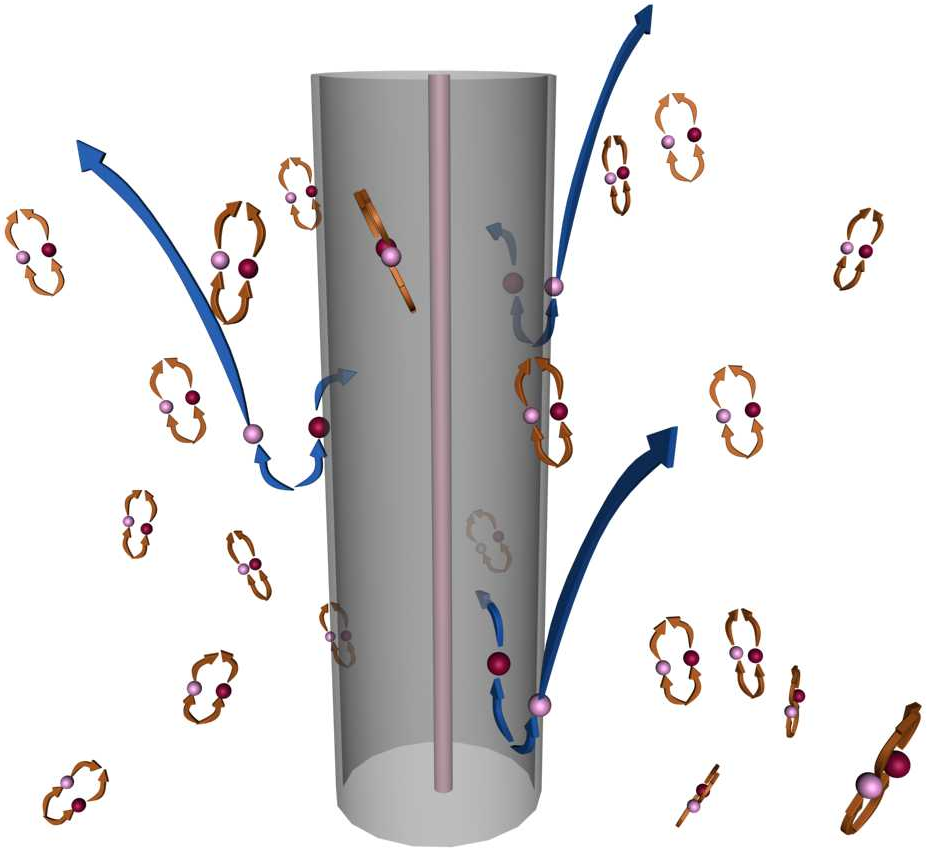
Entropy of a black hole is equal to ¼ the area measured in the units of Planck area.



Bits in a black hole?

Hawking Radiation

large momentum $mc \rightarrow$ particle creation



Black holes are not black!

They radiate due to particle creation/annihilation across the black hole horizon (this is a fudge, but…)

This radiation causes a black hole to lose mass

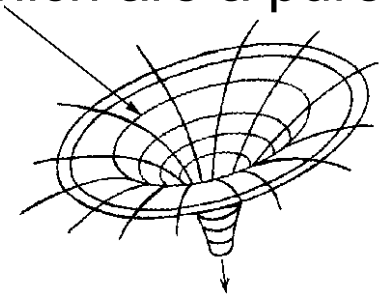
Black holes can evaporate!

http://library.thinkquest.org/C0126626/fate

No hair implies radiation should depend only on M, Q, L

Black Hole Information Paradox

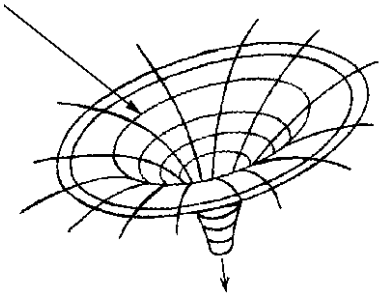
Throw qubit into a black hole (more properly state with initial conditions which are a pure state)



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Radiation doesn't depend on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ only on mass, charge, and angular momentum content

Black hole evaporates:



Where did the qubit go to? $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

Unitary evolution requires qubit should reappear somewhere…

This is the black hole information paradox

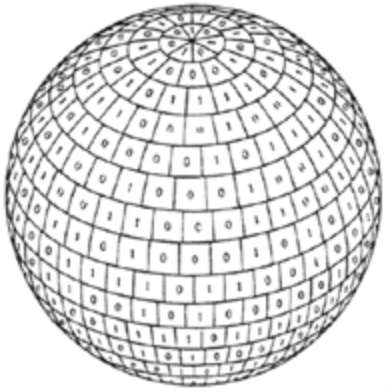
Black Hole Information Paradox



Whereas Stephen Hawking and Kip Thorne firmly believe that information swallowed by a black hole is forever hidden from the outside universe, and can never be revealed even as the black hole evaporates and completely disappears,
And whereas John Preskill firmly believes that a mechanism for the information to be released by the evaporating black hole must and will be found in the correct theory of quantum gravity,
Therefore Preskill offers, and Hawking/Thorne accept, a wager that:
When an initial pure quantum state undergoes gravitational collapse to form a black hole, the final state at the end of black hole evaporation will always be a pure quantum state.
The loser(s) will reward the winner(s) with an encyclopedia of the winner's choice, from which information can be recovered at will.
Stephen W. Hawking, Kip S. Thorne, John P. Preskill
Pasadena, California, 6 February 1997

Holographic Principle

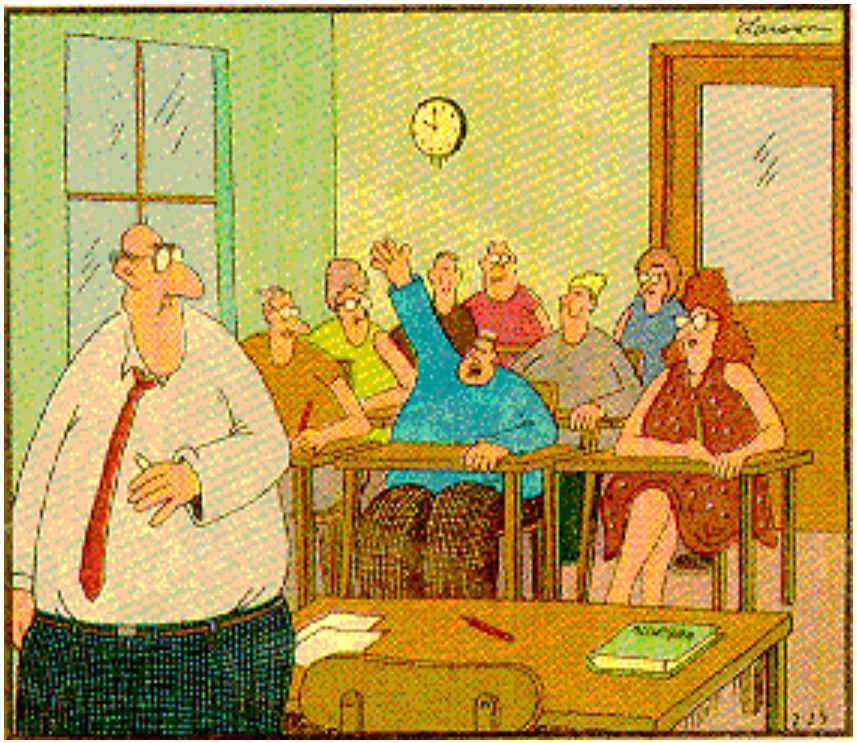
t'Hooft, Susskind: all of the information contained in a volume of space can be represented by a theory that lives in the boundary of that region



Side result: The ultimate limit to the storage of information is that if you try to pack more and more information onto your hard drive, then eventually this hard drive will collapse into a black hole.
What this information storage capacity of a hard drive?

$$\frac{1}{L_P^2} = \frac{c^3}{G\hbar} \approx 10^{66}\ \mathrm{cm}^{-2}$$

that's a lot of bits!



“Dave, may I be excused? My brain is full.”